

Assignment 2

My repo URL is: <https://github.com/AyerDaniel/turtlebot-maze>

This assignment is mostly contained within `object_detection_events/` folder.

This startup script should outline the mechanisms of the system:

```
$ object_detection_events-start.sh
1  #!/bin/bash
2
3  # Head into working directory.
4  cd /home/johnsmith/Desktop/njit/workspaces/turtlebot-maze
5
6  # Run containers. -d to run headless.
7  docker compose up -d zenoh-router
8  docker compose up -d zenoh-bridge
9  docker compose up -d demo-world-enhanced
10 docker compose up -d detector
11
12 # Move to project folder.
13 cd /home/johnsmith/Desktop/njit/workspaces/turtlebot-maze/object_detection_events
14
15 # Run containers.
16 docker compose up -d postgres
17 docker compose up object_detection_events
18
19 ## You should now see detections made by the object_detection_events container in its terminal output.
20 ## You should now see entries made into the postgres database.
```

Deliverables:

Detector

The detector used was the supplied detector container. The professor said we could use all the demo-* containers and their published Zenoh topics.

This can be found in: `detector/object_detector.py`.

Zenoh Publisher Logic

The publisher logic is as follows: A detection is made by the detector container and is published to Zenoh as JSON to topic: `maze/{robot_id}/detections/v1/{event_id}`.

This code can be found in: `object_detection_events/object_detection_events.py`.

Zenoh Ingest Worker

The worker subscribes to `maze/**/detections/v1/*`, parses the topic, and writes to a PostgreSQL database `detections_db`. I sued a docker container pulldown for postgres to create the db. The supplied schema createion SQL was used to create the schema and the Zenoh topic JSON.

This code can be found in: `object_detection_events/object_detection_events.py`.

world_assets.md

This file is located at: `object_detection_events/world_assets.md`.

Histogram

A sort of 'brute' script was written to visualize the output of the postgres detections table to create the histogram. Due to time constraints the workflow was:

1) Export detections table to csv.

2) Open in Libre Office Calc (Ubuntu Excel Option).

3) The column for class was copied and used to create a script. It is a hard coded list. This file does not dynamically generate the histogram from the postgres table. This is a refactor goal, but due to time constraints 'good enough' has prevailed.

This code can be found at: `plot_it.py`.

Reference:

Postgres Schema:

```
CREATE TABLE detection_events (  
  event_id uuid PRIMARY KEY,  
  run_id uuid NOT NULL,  
  robot_id text NOT NULL,  
  sequence bigint NOT NULL,  
  stamp timestampz NOT NULL,  
  
  image_frame_id text,  
  image_sha256 text,  
  width int,  
  height int,  
  encoding text,  
  
  x double precision,  
  y double precision,  
  yaw double precision,  
  vx double precision,  
  vy double precision,  
  wz double precision,  
  
  tf_ok boolean,  
  t_base_camera double precision[],  
  
  raw_event jsonb NOT NULL,  
  
  UNIQUE(run_id, robot_id, sequence)  
);  
  
CREATE TABLE detections (  
  det_pk bigserial PRIMARY KEY,  
  event_id uuid REFERENCES detection_events(event_id) ON DELETE CASCADE,  
  det_id uuid,  
  class_id int,  
  class_name text,  
  confidence double precision,  
  x1 double precision,  
  y1 double precision,  
  x2 double precision,  
  y2 double precision,  
  UNIQUE(event_id, det_id)  
);
```

JSON Schema

Publish JSON with this structure:

```
{
  "schema": "maze.detection.v1",
  "event_id": "uuid",
  "run_id": "uuid",
  "robot_id": "tb3_sim",
  "sequence": 1023,
  "image": {
    "topic": "/camera/image_raw",
    "stamp": {"sec": 0, "nanosec": 0},
    "frame_id": "camera_link",
    "width": 640,
    "height": 480,
    "encoding": "rgb8",
    "sha256": "hex"
  },
  "odometry": {
    "topic": "/odom",
    "frame_id": "odom",
    "x": 1.2,
    "y": 3.4,
    "yaw": 0.5,
    "vx": 0.1,
    "vy": 0.0,
    "wz": 0.02
  },
  "tf": {
    "base_frame": "base_footprint",
    "camera_frame": "camera_link",
    "t_base_camera": [16],
    "tf_ok": true
  },
  "detections": [
    {
      "det_id": "uuid",
      "class_id": 41,
      "class_name": "cup",
      "confidence": 0.91,
      "bbox_xyxy": [x1,y1,x2,y2]
    }
  ]
}
```

Report with counts:

	A	B	C	D	E	F	G	H	I	J
1	det_pk	event_id	det_id	class_id	class_name	confidence	x1	y1	x2	y2
2	2273	de7dd9b5-a378-423a-ac27-065aa02f6405	c3a6cbef-423a-4281-a9a1-9fb2f6df1e1b	28	suitcase	0.757	246	146	320	209
3	2274	de7dd9b5-a378-423a-ac27-065aa02f6405	1b2a37ab-4a4f-4cd3-b952-08d9d330d957	28	suitcase	0.815	245	146	320	209
4	2275	de7dd9b5-a378-423a-ac27-065aa02f6405	d3db74f3-f7d1-4d15-9ff4-e09489b00c8b	28	suitcase	0.715	237	145	320	208
5	2276	de7dd9b5-a378-423a-ac27-065aa02f6405	840f3ee0-12cd-4487-b614-659ee655cd79	28	suitcase	0.809	224	144	320	206
6	2277	de7dd9b5-a378-423a-ac27-065aa02f6405	8eb05b8d-b7fd-4046-a542-04ffe13c23fe	28	suitcase	0.69	204	143	320	202
7	2278	de7dd9b5-a378-423a-ac27-065aa02f6405	81a7ae09-f7d1-406d-8e88-087e63d13ff0	14	bird	0.529	229	77	257	126
8	2279	de7dd9b5-a378-423a-ac27-065aa02f6405	50fbbb51-4f7a-4cc5-8fb2-d2cec5858119	59	bed	0.559	0	146	320	240
9	2280	de7dd9b5-a378-423a-ac27-065aa02f6405	f465635f-3c89-4910-a3cc-262a5a29cf2f	59	bed	0.624	0	146	320	239
10	2281	de7dd9b5-a378-423a-ac27-065aa02f6405	3b3378e3-a04c-47fa-b880-34775af00f2e	59	bed	0.568	1	134	311	240
11	2282	de7dd9b5-a378-423a-ac27-065aa02f6405	cd7d45d3-26ea-42ed-b55d-a7155df07f22	28	suitcase	0.58	220	62	319	98
12	2283	de7dd9b5-a378-423a-ac27-065aa02f6405	ac1edd1d-e586-44a1-81d8-bc79bbeb876f	59	bed	0.622	1	151	294	239
13	2284	de7dd9b5-a378-423a-ac27-065aa02f6405	30d5bdad-37f9-4cfe-9cca-7eb01086c311	59	bed	0.561	0	148	320	239
14	2285	de7dd9b5-a378-423a-ac27-065aa02f6405	bf32f047-e0b1-4345-bbef-033365070579	59	bed	0.632	0	146	318	239
15	2286	de7dd9b5-a378-423a-ac27-065aa02f6405	d06a51a7-c6a7-4831-9e55-572b58e9035b	59	bed	0.627	0	146	318	239
16	2287	de7dd9b5-a378-423a-ac27-065aa02f6405	c515286c-4288-400b-b72c-d3ce7f9a3ac2	59	bed	0.665	0	151	320	240
17	2288	de7dd9b5-a378-423a-ac27-065aa02f6405	c54aebb8-4e25-4a0d-8a0d-349b0a50b4c9	59	bed	0.704	0	149	318	240
18	2289	de7dd9b5-a378-423a-ac27-065aa02f6405	b3016827-1ea6-4ad8-ad3d-736d7d28b594	59	bed	0.663	0	149	318	240
19	2290	de7dd9b5-a378-423a-ac27-065aa02f6405	5529d599-025a-4502-9547-d3bb4231ab73	59	bed	0.603	0	149	318	240
20	2291	de7dd9b5-a378-423a-ac27-065aa02f6405	0e58b931-d412-4cfd-b4df-ff99089f57a9	59	bed	0.581	0	150	318	240
21	2292	de7dd9b5-a378-423a-ac27-065aa02f6405	e40328d8-870c-486d-b55c-0a0aa84fcb1a	59	bed	0.581	1	145	315	240
22	2293	de7dd9b5-a378-423a-ac27-065aa02f6405	2d559a48-37d6-455a-891a-409d46b25043	59	bed	0.581	0	141	316	240
23	2294	de7dd9b5-a378-423a-ac27-065aa02f6405	2ba04013-eda6-4c90-856f-31bfff6a595c	59	bed	0.509	1	141	317	240
24	2295	de7dd9b5-a378-423a-ac27-065aa02f6405	5ac611bd-a028-401a-b4ba-5b3493e3f8a8	59	bed	0.603	1	138	306	240
25	2296	de7dd9b5-a378-423a-ac27-065aa02f6405	ed5175b8-f538-4803-818f-76fea9d09f5e	28	suitcase	0.591	269	0	320	47
26	2297	de7dd9b5-a378-423a-ac27-065aa02f6405	e6e02ce5-7e15-48d2-be28-db051f7cd8d9	28	suitcase	0.51	2	0	319	90
27	2298	de7dd9b5-a378-423a-ac27-065aa02f6405	7d20c975-ffa3-49f2-afe2-7aa5c96d5fb2	59	bed	0.547	0	170	320	240
28	2299	de7dd9b5-a378-423a-ac27-065aa02f6405	d2417d28-a62d-4413-b2b3-3cbc694f6f39	32	sports ball	0.662	291	128	320	155
29	2300	de7dd9b5-a378-423a-ac27-065aa02f6405	d55ea75b-3647-4a1c-bc25-aa299828b452	32	sports ball	0.828	254	127	281	152
30	2301	de7dd9b5-a378-423a-ac27-065aa02f6405	de828e85-cfc7-4aa1-afab-08861d43f0cc	32	sports ball	0.581	224	127	250	152
31	2302	de7dd9b5-a378-423a-ac27-065aa02f6405	4a09595a-2e09-4724-83b4-9b9d8f096436	32	sports ball	0.77	208	127	232	151
32	2303	de7dd9b5-a378-423a-ac27-065aa02f6405	146f2e1d-cacb-4b91-a508-42af2340e66c	32	sports ball	0.823	208	127	232	151
33	2304	de7dd9b5-a378-423a-ac27-065aa02f6405	776a331d-e1f0-4d01-b8c7-797f8e7ce20b	32	sports ball	0.833	208	127	232	151
34	2305	de7dd9b5-a378-423a-ac27-065aa02f6405	a0f41338-05ec-4e2d-a06c-dd6822f6f12a	32	sports ball	0.83	208	127	232	151
35										
36		suitcase		8						
37		bird		1						
38		bed		17						
39		sports ball		7						

Histogram

